Variance and Concentration

COMP2711 Discrete Mathematical Tools

Mingxun Zhou (mingxunz@ust.hk)

Review: Expectation

Expectation $\mathbb{E}[X] = \sum_x x \Pr[X = x]$

Functions of X $\mathbb{E}[g(X)] = \sum_x g(x) \Pr[X = x]$

Linearity $\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$

Linearity does not require independence.

In general, $\mathbb{E}[X^2] \neq (\mathbb{E}[X])^2$.

Example: Two games

The net gain, in dollars:

	Possible gains	Probabilities
Game X	$-\$2$ or $+\$4$	$1/2$ each
Game Y	$-\$98$ or $+\$100$	$1/2$ each

What is the expected gain of each game?

Example: Two games

The net gain, in dollars:

	Possible gains	Probabilities
Game X	$-\$2$ or $+\$4$	$1/2$ each
Game Y	$-\$98$ or $+\$100$	$1/2$ each

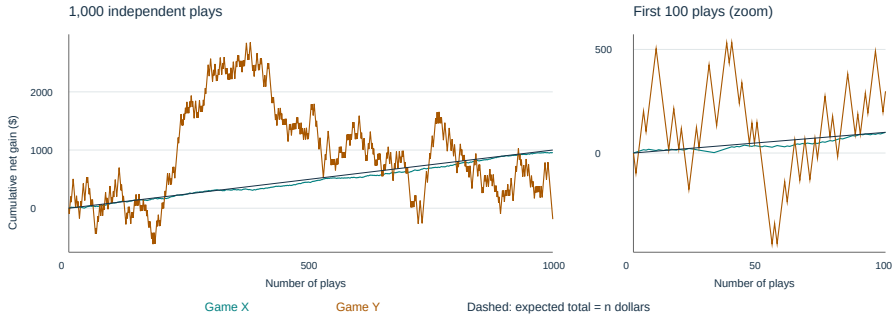
What is the expected gain of each game?

$$\mathbb{E}[X] = \tfrac{1}{2}(-2) + \tfrac{1}{2}(4) = 1, \qquad \mathbb{E}[Y] = \tfrac{1}{2}(-98) + \tfrac{1}{2}(100) = 1.$$

Same expected gain. Are the games equally predictable?

Two games: 1,000 plays

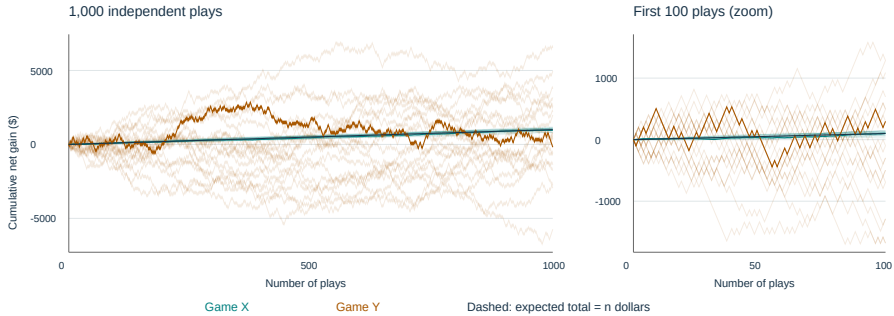
Play each game independently 1,000 times, starting from $0.



Expected total after n plays: n dollars for either game.

Two games: 1,000 plays

Play each game independently 1,000 times, starting from $0.



20 more runs per game; vertical axes rescaled to fit.

Three main concepts

1. Variance	How much do results vary around their mean?
2. Concentration	How unlikely are unusually large values or deviations?
3. Independent averages	Why does averaging make an estimate more reliable?

Measure spread $\longrightarrow$ bound error $\longrightarrow$ explain reliable averages.

Deviation and concentration

How far can a random result be from its average?

How likely is it to stay close?

These questions underlie risk assessment, error analysis, and many other practical problems.

SECTION 1

Variance

Variance

MUST UNDERSTAND

Let $\mu = \mathbb{E}[X]$. The variance of X is

$$\mathbf{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

The average squared distance from the mean.

For discrete X: $\mathrm{Var}(X) = \sum_x (x - \mu)^2 \Pr[X = x]$.

Variance

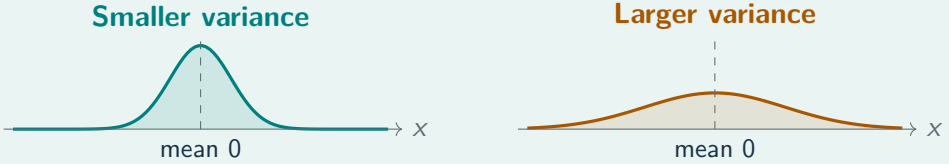
MUST UNDERSTAND

Let $\mu = \mathbb{E}[X]$. The variance of X is

$$\mathbf{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

The average squared distance from the mean.

For discrete X: $\mathrm{Var}(X) = \sum_x (x - \mu)^2 \Pr[X = x]$.



Normal distributions, shown on the same scales.

Two games: Variance

Definition: $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$

Game X

Gain x	-2	4
Probability	$1/2$	$1/2$

$\mathbb{E}[X] = \frac{1}{2}(-2) + \frac{1}{2}(4) = 1$

Game Y

Gain y	-98	100
Probability	$1/2$	$1/2$

$\mathbb{E}[Y] = \frac{1}{2}(-98) + \frac{1}{2}(100) = 1$

Two games: Variance

Definition: $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$

Game X		
Gain x	-2	4
Probability	$1/2$	$1/2$
$\mathbb{E}[X] = \frac{1}{2}(-2) + \frac{1}{2}(4) = 1$		
Deviation $x - 1$	-3	3

Game Y		
Gain y	-98	100
Probability	$1/2$	$1/2$
$\mathbb{E}[Y] = \frac{1}{2}(-98) + \frac{1}{2}(100) = 1$		

Two games: Variance

Definition: $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$

Game X

Gain x	-2	4
Probability	$1/2$	$1/2$

$\mathbb{E}[X] = \frac{1}{2}(-2) + \frac{1}{2}(4) = 1$

Deviation $x - 1$	-3	3
Squared deviation	9	9

Game Y

Gain y	-98	100
Probability	$1/2$	$1/2$

$\mathbb{E}[Y] = \frac{1}{2}(-98) + \frac{1}{2}(100) = 1$

Two games: Variance

Definition: $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$

Game X		
Gain x	-2	4
Probability	$1/2$	$1/2$
$\mathbb{E}[X] = \frac{1}{2}(-2) + \frac{1}{2}(4) = 1$		
Deviation $x - 1$	-3	3
Squared deviation	9	9
$\mathrm{Var}(X) = \frac{1}{2}(9) + \frac{1}{2}(9) = \mathbf{9}$		

Game Y		
Gain y	-98	100
Probability	$1/2$	$1/2$
$\mathbb{E}[Y] = \frac{1}{2}(-98) + \frac{1}{2}(100) = 1$		

Two games: Variance

Definition: $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$

Game X		
Gain x	-2	4
Probability	$1/2$	$1/2$
$\mathbb{E}[X] = \frac{1}{2}(-2) + \frac{1}{2}(4) = 1$		
Deviation $x - 1$	-3	3
Squared deviation	9	9
$\mathrm{Var}(X) = \frac{1}{2}(9) + \frac{1}{2}(9) = \mathbf{9}$		

Game Y		
Gain y	-98	100
Probability	$1/2$	$1/2$
$\mathbb{E}[Y] = \frac{1}{2}(-98) + \frac{1}{2}(100) = 1$		
Deviation $y - 1$	-99	99
Squared deviation	9801	9801
$\mathrm{Var}(Y) = \frac{1}{2}(9801) + \frac{1}{2}(9801) = 9801$		

Standard deviation

The standard deviation is

$$\sigma = \sqrt{\text{Var}(X)}.$$

	Mean	Variance	Standard deviation
Game X	\$1	9 dollars^2	\$3
Game Y	\$1	9801 dollars^2	\$99

Standard deviation has the same units as X.

A common equivalent form of variance

Write $\mu = \mathbb{E}[X]$.

$$\mathrm{Var}(X) = \underbrace{\mathbb{E}[(X-\mu)^2]}_{\text{definition}} = \underbrace{\mathbb{E}[X^2] - \mu^2}_{\text{equivalent form}}.$$

A common equivalent form of variance

Write $\mu = \mathbb{E}[X]$.

$$\mathrm{Var}(X) = \underbrace{\mathbb{E}[(X-\mu)^2]}_{\text{definition}} = \underbrace{\mathbb{E}[X^2] - \mu^2}_{\text{equivalent form}}.$$

Proof.

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[(X-\mu)^2] \\ &= \mathbb{E}[X^2 - 2\mu X + \mu^2]\end{aligned}$$

A common equivalent form of variance

Write $\mu = \mathbb{E}[X]$.

$$\mathrm{Var}(X) = \underbrace{\mathbb{E}[(X-\mu)^2]}_{\text{definition}} = \underbrace{\mathbb{E}[X^2] - \mu^2}_{\text{equivalent form}}.$$

Proof.

$$\begin{aligned}
\mathrm{Var}(X) &= \mathbb{E}[(X-\mu)^2] \\
&= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\
&= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \qquad \text{(linearity of expectation)}
\end{aligned}$$

A common equivalent form of variance

Write $\mu = \mathbb{E}[X]$.

$$\mathrm{Var}(X) = \underbrace{\mathbb{E}[(X-\mu)^2]}_{\text{definition}} = \underbrace{\mathbb{E}[X^2] - \mu^2}_{\text{equivalent form}}.$$

Proof.

$$\begin{aligned}
\mathrm{Var}(X) &= \mathbb{E}[(X-\mu)^2] \\
&= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\
&= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \qquad \text{(linearity of expectation)} \\
&= \mathbb{E}[X^2] - 2\mu \cdot \mu + \mu^2 \qquad \text{(substitute } \mathbb{E}[X] = \mu\text{)}
\end{aligned}$$

A common equivalent form of variance

Write $\mu = \mathbb{E}[X]$.

$$\mathrm{Var}(X) = \underbrace{\mathbb{E}[(X-\mu)^2]}_{\text{definition}} = \underbrace{\mathbb{E}[X^2] - \mu^2}_{\text{equivalent form}}.$$

Proof.

$$\begin{aligned}
\mathrm{Var}(X) &= \mathbb{E}[(X-\mu)^2] \\
&= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\
&= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \qquad \text{(linearity of expectation)} \\
&= \mathbb{E}[X^2] - 2\mu \cdot \mu + \mu^2 \qquad \text{(substitute } \mathbb{E}[X] = \mu\text{)} \\
&= \mathbb{E}[X^2] - \mu^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.
\end{aligned}$$

Example: A fair die

$$\mu = \mathbb{E}[X]$$
$$\text{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - \mu^2$$

Roll a fair six-sided die. Let D be the number shown.

Find $\text{Var}(D)$**.**

Example: A fair die

$\mu = \mathbb{E}[X]$
$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - \mu^2$

Roll a fair six-sided die. Let D be the number shown.

Find $\mathrm{Var}(D)$**.**

$$\mathbb{E}[D] = \frac{1+2+3+4+5+6}{6} = \frac{7}{2}, \qquad \mathbb{E}[D^2] = \frac{1+4+9+16+25+36}{6} = \frac{91}{6}.$$

Example: A fair die

$\mu = \mathbb{E}[X]$
$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - \mu^2$

Roll a fair six-sided die. Let D be the number shown.

Find $\mathrm{Var}(D)$**.**

$$\mathbb{E}[D] = \frac{1+2+3+4+5+6}{6} = \frac{7}{2}, \qquad \mathbb{E}[D^2] = \frac{1+4+9+16+25+36}{6} = \frac{91}{6}.$$

$$\mathrm{Var}(D) = \mathbb{E}[D^2] - (\mathbb{E}[D])^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}.$$

Shifting and scaling

For constants $a, b \in \mathbb{R}$,

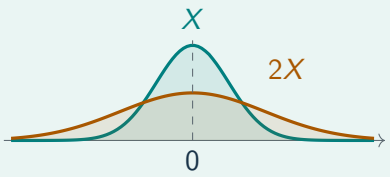
$$\mathrm{Var}(aX) = a^2 \,\mathrm{Var}(X), \qquad \mathrm{Var}(X + b) = \mathrm{Var}(X).$$

Shifting and scaling

For constants $a, b \in \mathbb{R}$,

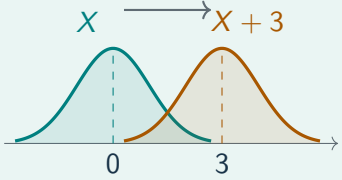
$$\mathrm{Var}(aX) = a^2 \,\mathrm{Var}(X), \qquad \mathrm{Var}(X + b) = \mathrm{Var}(X).$$

Scaling: $X \longrightarrow 2X$



$\mathrm{Var}(2X) = 4\,\mathrm{Var}(X)$

Shifting: $X \longrightarrow X + 3$



$\mathrm{Var}(X + 3) = \mathrm{Var}(X)$

Exercise: Try to prove them by yourself. Just follow the definition.

Review: Linearity of expectation

For any two random variables X, Y with finite expectations,

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

Regardless of whether X and Y are independent.

Review: Linearity of expectation

For any two random variables X, Y with finite expectations,

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

Regardless of whether X and Y are independent.

Does the same rule hold for variance?

$$\mathrm{Var}(X + Y) \stackrel{?}{=} \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Variance of a sum

Assume X and Y have finite variances.

$$\text{Var}(X+Y) = \mathbb{E}[(X+Y)^2] - (\mathbb{E}[X] + \mathbb{E}[Y])^2$$

Variance of a sum

Assume X and Y have finite variances.

$$\begin{aligned}\mathrm{Var}(X+Y) &= \mathbb{E}[(X+Y)^2] - (\mathbb{E}[X]+\mathbb{E}[Y])^2 \\ &= \mathbb{E}[X^2 + 2XY + Y^2] \\ &\quad - \left((\mathbb{E}[X])^2 + 2\mathbb{E}[X]\mathbb{E}[Y] + (\mathbb{E}[Y])^2\right)\end{aligned}$$

Expand both squares using $(a+b)^2 = a^2 + 2ab + b^2$.

Variance of a sum

Assume X and Y have finite variances.

$$\begin{aligned}\mathrm{Var}(X+Y) &= \mathbb{E}[(X+Y)^2] - (\mathbb{E}[X]+\mathbb{E}[Y])^2 \\ &= \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] \\ &\quad - ((\mathbb{E}[X])^2 + 2\mathbb{E}[X]\mathbb{E}[Y] + (\mathbb{E}[Y])^2)\end{aligned}$$

Apply linearity of expectation to the first term.

Variance of a sum

Assume X and Y have finite variances.

$$\begin{aligned}
\mathrm{Var}(X+Y) &= \mathbb{E}[(X+Y)^2] - (\mathbb{E}[X]+\mathbb{E}[Y])^2 \\
&= \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] \\
&\quad - ((\mathbb{E}[X])^2 + 2\mathbb{E}[X]\mathbb{E}[Y] + (\mathbb{E}[Y])^2) \\
&= \underbrace{(\mathbb{E}[X^2] - (\mathbb{E}[X])^2)}_{\mathrm{Var}(X)} + \underbrace{(\mathbb{E}[Y^2] - (\mathbb{E}[Y])^2)}_{\mathrm{Var}(Y)} \\
&\quad + 2\underbrace{(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y])}_{\text{covariance } \mathrm{Cov}(X,Y)}.
\end{aligned}$$

Variance of a sum

Assume X and Y have finite variances.

$$\begin{aligned}
\mathrm{Var}(X+Y) &= \mathbb{E}[(X+Y)^2] - (\mathbb{E}[X]+\mathbb{E}[Y])^2 \\
&= \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] \\
&\quad - \left((\mathbb{E}[X])^2 + 2\mathbb{E}[X]\mathbb{E}[Y] + (\mathbb{E}[Y])^2\right) \\
&= \mathrm{Var}(X) + \mathrm{Var}(Y) \\
&\quad + 2\underbrace{\left(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]\right)}_{\text{covariance } \mathrm{Cov}(X,Y)}.
\end{aligned}$$

Variance of a sum

Assume X and Y have finite variances.

$$\begin{aligned}\mathrm{Var}(X+Y) &= \mathbb{E}[(X+Y)^2] - (\mathbb{E}[X]+\mathbb{E}[Y])^2 \\ &= \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] \\ &\quad - \left((\mathbb{E}[X])^2 + 2\mathbb{E}[X]\mathbb{E}[Y] + (\mathbb{E}[Y])^2\right) \\ &= \mathrm{Var}(X) + \mathrm{Var}(Y) \\ &\quad + 2\underbrace{\left(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]\right)}_{\text{covariance } \mathrm{Cov}(X,Y)}.\end{aligned}$$

In general, $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$.

Variance is therefore **not additive in general**.

Linearity of variance under independence

If X and Y are **independent** and have finite variances,

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Linearity of variance under independence

If X and Y are **independent** and have finite variances,

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Independence gives $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

(Exercise: prove this by yourself.)

Hence,

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0.$$

Linearity of variance under independence

If X and Y are **independent** and have finite variances,

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Independence gives $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

(Exercise: prove this by yourself.)

Hence,

$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0.$$

More generally, for **independent** $X_1, \ldots, X_n$ with finite variances (pairwise independence is enough),

$$\mathrm{Var}(X_1 + \cdots + X_n) = \mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n).$$

Case study: Variance of Binomial Distribution

Review: Binomial distribution

$H \sim \text{Bin}(n, p)$ counts Yes results in n **independent** trials, each with Yes probability p.

Let $I_j = 1$ if trial j says Yes, and $I_j = 0$ otherwise.

$$H = I_1 + \cdots + I_n.$$

Case study: Variance of Binomial Distribution

Review: Binomial distribution

$H \sim \text{Bin}(n, p)$ counts Yes results in n **independent** trials, each with Yes probability p.

Let $I_j = 1$ if trial j says Yes, and $I_j = 0$ otherwise.

$$H = I_1 + \cdots + I_n.$$

What is $\text{Var}(H)$?

Case study: Variance of Binomial Distribution

Trial j says Yes with probability p, and No with probability $1 - p$.

$$I_j = \begin{cases} 1, & \text{trial } j \text{ says Yes,} \\ 0, & \text{trial } j \text{ says No.} \end{cases}$$

I_j	0	1
I_j^2	0	1
Probability	$1 - p$	p

Case study: Variance of Binomial Distribution

Trial j says Yes with probability p, and No with probability $1 - p$.

$$I_j = \begin{cases} 1, & \text{trial } j \text{ says Yes,} \\ 0, & \text{trial } j \text{ says No.} \end{cases}$$

I_j	0	1
I_j^2	0	1
Probability	$1-p$	p

$$\mathbb{E}[I_j^2] = \mathbb{E}[I_j] = p.$$

$$\mathrm{Var}(I_j) = \mathbb{E}[I_j^2] - (\mathbb{E}[I_j])^2 = p - p^2 = p(1-p).$$

Case study: Variance of Binomial Distribution

$H \sim \text{Bin}(n, p)$, with $H = I_1 + \cdots + I_n$.

$\mathbb{E}[I_j] = p$. By **linearity of expectation**,

$$\mathbb{E}[H] = \mathbb{E}[I_1] + \cdots + \mathbb{E}[I_n] = np.$$

Case study: Variance of Binomial Distribution

$H \sim \mathrm{Bin}(n, p)$, with $H = I_1 + \cdots + I_n$.

$\mathbb{E}[I_j] = p$. By **linearity of expectation**,

$$\mathbb{E}[H] = \mathbb{E}[I_1] + \cdots + \mathbb{E}[I_n] = np.$$

$\mathrm{Var}(I_j) = p(1-p)$. By **variance additivity under independence**,

$$\mathrm{Var}(H) = \mathrm{Var}(I_1) + \cdots + \mathrm{Var}(I_n) = np(1-p).$$

$$\mathbb{E}[H] = np, \qquad \mathrm{Var}(H) = np(1-p).$$

Case study: Variance of Binomial Distribution

$H \sim \mathrm{Bin}(n, p)$, with $H = I_1 + \cdots + I_n$.

$\mathbb{E}[I_j] = p$. By **linearity of expectation**,

$$\mathbb{E}[H] = \mathbb{E}[I_1] + \cdots + \mathbb{E}[I_n] = np.$$

$\mathrm{Var}(I_j) = p(1-p)$. By **variance additivity under independence**,

$$\mathrm{Var}(H) = \mathrm{Var}(I_1) + \cdots + \mathrm{Var}(I_n) = np(1-p).$$

$$\mathbb{E}[H] = np, \qquad \mathrm{Var}(H) = np(1-p).$$

Example: 100 independent fair flips:

$$\mathbb{E}[H] = 50, \qquad \mathrm{Var}(H) = 25, \qquad \sigma = \sqrt{\mathrm{Var}(H)} = 5.$$

SECTION 2

Concentration

Concentration: What and why?

MUST UNDERSTAND

How likely is a random result to be far from its average?

We want an upper bound on the chance of a large deviation.



Normal distribution illustration

Concentration: What and why?

MUST UNDERSTAND

How likely is a random result to be far from its average?

We want an upper bound on the chance of a large deviation.



Normal distribution illustration

Computing the exact probability from the full distribution can be too difficult.

A small upper bound tells us the result is likely to stay close to its average.

Example: Items that stay in place

Uniformly permute n items, where $n \geq 10$.

Let X count the items that remain in their original positions.

For each item i, let $I_i = 1$ if it stays in place, and 0 otherwise.

$$X = I_1 + \cdots + I_n, \qquad \Pr[I_i = 1] = \frac{1}{n}.$$

Example: Items that stay in place

Uniformly permute n items, where $n \geq 10$.

Let X count the items that remain in their original positions.

For each item i, let $I_i = 1$ if it stays in place, and 0 otherwise.

$$X = I_1 + \cdots + I_n, \qquad \Pr[I_i = 1] = \frac{1}{n}.$$

$$\mathbb{E}[X] = \mathbb{E}[I_1] + \cdots + \mathbb{E}[I_n] = n \cdot \frac{1}{n} = 1.$$

Example: Items that stay in place

Uniformly permute n items, where $n \geq 10$.

Let X count the items that remain in their original positions.

For each item i, let $I_i = 1$ if it stays in place, and 0 otherwise.

$$X = I_1 + \cdots + I_n, \qquad \Pr[I_i = 1] = \frac{1}{n}.$$

$$\mathbb{E}[X] = \mathbb{E}[I_1] + \cdots + \mathbb{E}[I_n] = n \cdot \frac{1}{n} = 1.$$

What upper bound can we give for $\Pr[X > 10]$?

Why not calculate the probability directly?

To calculate $\Pr[X > 10]$ exactly, we need to count **derangements**: arrangements in which none of the remaining items stays in place.

Why not calculate the probability directly?

To calculate $\Pr[X > 10]$ exactly, we need to count **derangements**: arrangements in which none of the remaining items stays in place.

However, knowing only $\mathbb{E}[X] = 1$, we can already conclude

$$\Pr[X > 10] < \frac{1}{10}.$$

How?

Markov's inequality

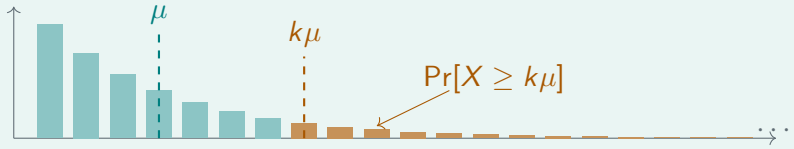
For a **nonnegative** random variable X, write $\mu = \mathbb{E}[X]$, with $0 < \mu < \infty$.

$$\Pr[X \geq k\mu] \leq \frac{1}{k} \qquad \text{for } k \geq 1.$$

Markov's inequality

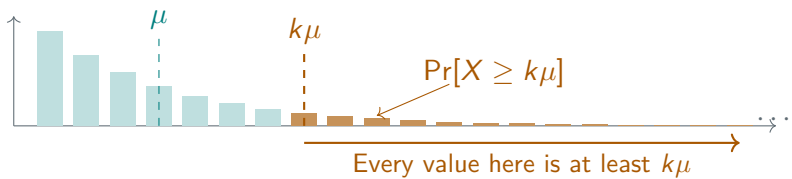
For a **nonnegative** random variable X, write $\mu = \mathbb{E}[X]$, with $0 < \mu < \infty$.

$$\Pr[X \geq k\mu] \leq \frac{1}{k} \qquad \text{for } k \geq 1.$$



Why Markov's inequality holds

$X \geq 0$, $\mu = \mathbb{E}[X] > 0$, and $k \geq 1$.

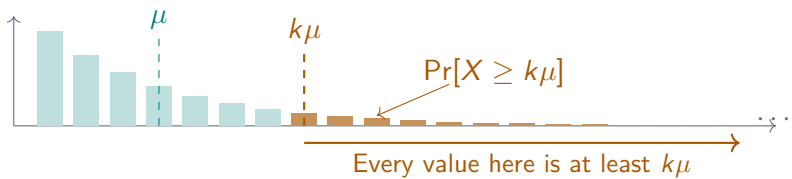


For the sake of contradiction, assume $\Pr[X \geq k\mu] > 1/k$.

The tail's **value** $\times$ **probability** alone would then exceed μ.

Why Markov's inequality holds

$X \geq 0$, $\mu = \mathbb{E}[X] > 0$, and $k \geq 1$.



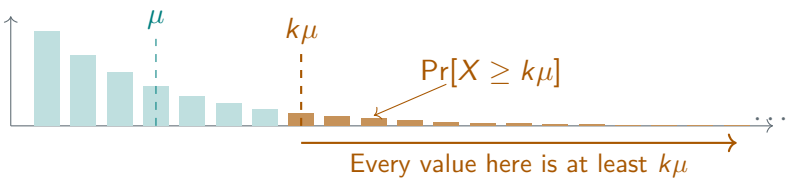
For the sake of contradiction, assume $\Pr[X \geq k\mu] > 1/k$.

The tail's **value** $\times$ **probability** alone would then exceed μ.

$$\mathbb{E}[X] \geq \underbrace{k\mu}_{\text{minimum tail value}} \underbrace{\Pr[X \geq k\mu]}_{\text{tail probability}} > k\mu \cdot \frac{1}{k} = \mu \quad \text{— impossible!}$$

Why Markov's inequality holds

$X \geq 0$, $\mu = \mathbb{E}[X] > 0$, and $k \geq 1$.



For the sake of contradiction, assume $\Pr[X \geq k\mu] > 1/k$.

The tail's **value** $\times$ **probability** alone would then exceed μ.

$$\Pr[X \geq k\mu] \leq \frac{1}{k}.$$

Markov requires a nonnegative random variable

For a **nonnegative** random variable X, write $\mu = \mathbb{E}[X]$, with $0 < \mu < \infty$.

$$\Pr[X \geq k\mu] \leq \frac{1}{k} \qquad \text{for } k \geq 1.$$

Without nonnegativity, negative value $\times$ probability terms can cancel positive ones.

Applying Markov to Variance!

Let $\mu = \mathbb{E}[X]$ and $0 < \sigma^2 = \mathrm{Var}(X) < \infty$.

The squared distance from the mean is a **nonnegative random variable**:

$$Z = (X - \mu)^2 \geq 0, \qquad \mathbb{E}[Z] = \sigma^2.$$

Applying Markov to Variance!

Let $\mu = \mathbb{E}[X]$ and $0 < \sigma^2 = \mathrm{Var}(X) < \infty$.

The squared distance from the mean is a **nonnegative random variable**:

$$Z = (X - \mu)^2 \geq 0, \qquad \mathbb{E}[Z] = \sigma^2.$$

For $t \geq 1$, a large deviation is the same event as a large squared distance:

$$\begin{aligned} \Pr[|X - \mu| \geq t\sigma] &= \Pr[(X - \mu)^2 \geq t^2\sigma^2] \\ &= \Pr[Z \geq t^2\mathbb{E}[Z]] \end{aligned}$$

Applying Markov to Variance!

Let $\mu = \mathbb{E}[X]$ and $0 < \sigma^2 = \mathrm{Var}(X) < \infty$.

The squared distance from the mean is a **nonnegative random variable**:

$$Z = (X - \mu)^2 \geq 0, \qquad \mathbb{E}[Z] = \sigma^2.$$

For $t \geq 1$, a large deviation is the same event as a large squared distance:

$$\begin{aligned} \Pr[|X - \mu| \geq t\sigma] &= \Pr[(X - \mu)^2 \geq t^2\sigma^2] \\ &= \Pr[Z \geq t^2\mathbb{E}[Z]] \\ &\leq \frac{1}{t^2} \qquad \text{(Markov with } k = t^2\text{)}. \end{aligned}$$

Chebyshev's inequality

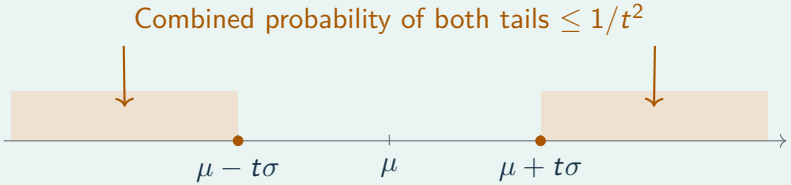
Let r.v. X have mean $\mu = \mathbb{E}[X]$ and $0 < \sigma^2 = \mathrm{Var}[X] < \infty$. For every $t \geq 1$,

$$\Pr[|X - \mu| \geq t\sigma] \leq \frac{1}{t^2}.$$

Chebyshev's inequality

Let r.v. X have mean $\mu = \mathbb{E}[X]$ and $0 < \sigma^2 = \mathrm{Var}[X] < \infty$. For every $t \geq 1$,

$$\Pr[|X - \mu| \geq t\sigma] \leq \frac{1}{t^2}.$$



Markov vs. Chebyshev

Markov: use the mean

For $X \geq 0$ with $\mu = \mathbb{E}[X] > 0$,

$$\Pr[X \geq k\mu] \leq \frac{1}{k}.$$

First-moment method

Chebyshev: use the variance

For X with mean μ and $\sigma^2 > 0$,

$$\Pr[|X - \mu| \geq t\sigma] \leq \frac{1}{t^2}.$$

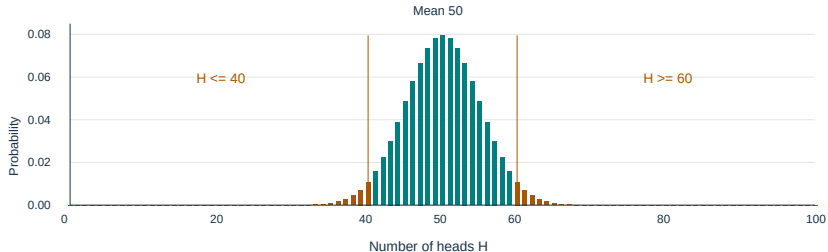
Second-moment method

The displayed means and variances are finite; $k, t \geq 1$.

Example: 100 fair flips

Flip a fair coin independently 100 times. Let H count heads.

$$\mathbb{E}[H] = 50, \qquad \mathrm{Var}(H) = 25.$$



How large can $\Pr[H \leq 40 \text{ or } H \geq 60]$ be?

100 fair flips: Calculation

$H \sim \text{Bin}(100, 1/2)$: mean 50, variance 25.

$$\{H \leq 40 \text{ or } H \geq 60\} = \{|H - 50| \geq 10\}.$$

100 fair flips: Calculation

$H \sim \text{Bin}(100, 1/2)$: mean 50, variance 25.

$$\{H \leq 40 \text{ or } H \geq 60\} = \{|H - 50| \geq 10\}.$$

$$\Pr[|H - 50| \geq 10] = \Pr[|H - 50| \geq 2\sigma] \leq \frac{1}{2^2} = \frac{1}{4}, \qquad \sigma = 5.$$

100 fair flips: Calculation

$H \sim \text{Bin}(100, 1/2)$: mean 50, variance 25.

$$\{H \leq 40 \text{ or } H \geq 60\} = \{|H - 50| \geq 10\}.$$

$$\Pr[|H - 50| \geq 10] = \Pr[|H - 50| \geq 2\sigma] \leq \frac{1}{2^2} = \frac{1}{4}, \qquad \sigma = 5.$$

Therefore,

$$\Pr[H \leq 40 \text{ or } H \geq 60] \leq \frac{1}{4}.$$

SECTION 3

Independent averages and the law of large numbers

Recall: Expectation as a long-run average

We introduced expectation with this intuition:

The expectation of a random variable X is its long-run average.

Recall: Expectation as a long-run average

We introduced expectation with this intuition:

The expectation of a random variable X is its long-run average.

Why should the average of many independent observations approach $\mathbb{E}[X]$?

Independent averages

MUST UNDERSTAND

Independent $X_1, \ldots, X_n$ each have mean μ and finite variance σ^2.

$$\overline{X}_n = \frac{X_1 + \cdots + X_n}{n}.$$

Independent averages

MUST UNDERSTAND

Independent $X_1, \ldots, X_n$ each have mean μ and finite variance σ^2.

$$\overline{X}_n = \frac{X_1 + \cdots + X_n}{n}.$$

$$\mathbb{E}[\overline{X}_n] = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}[X_i] = \mu.$$

Independent averages

MUST UNDERSTAND

Independent $X_1, \ldots, X_n$ each have mean μ and finite variance σ^2.

$$\overline{X}_n = \frac{X_1 + \cdots + X_n}{n}.$$

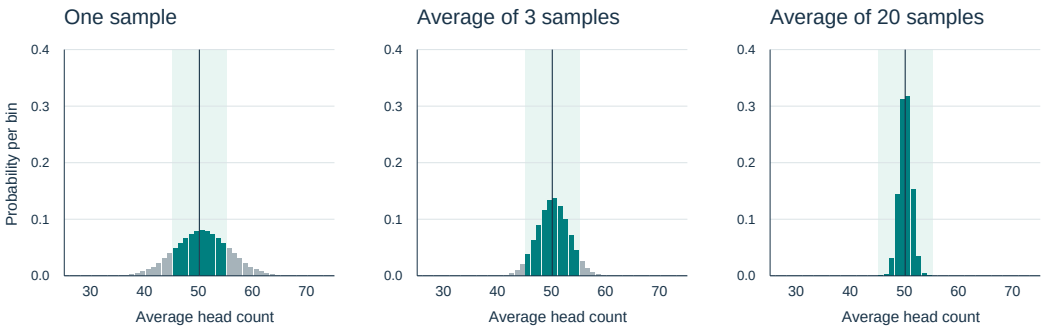
$$\mathbb{E}[\overline{X}_n] = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}[X_i] = \mu.$$

$$\mathrm{Var}(\overline{X}_n) = \underbrace{\frac{1}{n^2}}_{\text{scaling}} \underbrace{\sum_{i=1}^{n} \mathrm{Var}(X_i)}_{\text{independent sum}} = \frac{\sigma^2}{n}.$$

The mean stays the same; the variance shrinks by a factor of n.

More samples: A more concentrated average

Independent samples from $\text{Bin}(100, 1/2)$.



The law of large numbers

Recall Chebyshev: $\Pr[|X - \mu| \geq t\sigma] \leq 1/t^2,\ t \geq 1.$

$X_1, X_2, \ldots$: independent samples from the same distribution, with finite mean μ and variance $\sigma^2 > 0$.

$$\overline{X}_n = \frac{X_1 + \cdots + X_n}{n}, \qquad \mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}.$$

The law of large numbers

Recall Chebyshev: $\Pr[|X - \mu| \geq t\sigma] \leq 1/t^2,\ t \geq 1.$

$X_1, X_2, \ldots$: independent samples from the same distribution, with finite mean μ and variance $\sigma^2 > 0$.

$$\overline{X}_n = \frac{X_1 + \cdots + X_n}{n}, \qquad \mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}.$$

For fixed $\varepsilon > 0$, apply Chebyshev to $\overline{X}_n$, whose standard deviation is $\sigma/\sqrt{n}$, with $t = \varepsilon\sqrt{n}/\sigma \geq 1$ for large n:

$$\Pr[|\overline{X}_n - \mu| \geq \varepsilon] \leq \frac{1}{(\varepsilon\sqrt{n}/\sigma)^2} = \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0.$$

The probability of a fixed-size error tends to zero as n grows.

What the law does not say

A fair coin has just produced ten heads.

Does the next flip become more likely to be tails?

What the law does not say

A fair coin has just produced ten heads.

Does the next flip become more likely to be tails?

$$\Pr[\text{next flip is heads} \mid \text{first ten are heads}] = \frac{1}{2}.$$

Independent flips do not compensate for the past.

The average can move farther from the mean after another observation.

The law concerns the probability of error as n grows.

Summary

Variance	$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - \mu^2$
Independent sums	$\mathrm{Var}(\sum_i X_i) = \sum_i \mathrm{Var}(X_i)$
Markov: $X \geq 0$	$\Pr[X \geq a] \leq \mathbb{E}[X]/a, \quad a > 0$
Chebyshev	$\Pr[\lvert X-\mu\rvert \geq t] \leq \sigma^2/t^2, \quad t > 0$
Independent averages	$\mathrm{Var}(\overline{X}_n) = \sigma^2/n$

Here $\mu = \mathbb{E}[X]$ and $\sigma^2 = \mathrm{Var}(X) < \infty$. For averages, the variables share this mean and variance.

Practice: Variance and Markov

1. Two independent fair flips: X counts heads. Find $\mathbb{E}[X^2]$, $\mathrm{Var}(X)$, and $\mathrm{Var}(3X - 2)$.
2. Independent X, Y have means 3 and 5, variances 4 and 9. Find the mean and variance of $2X - Y + 1$. Which calculation survives without independence?
3. $Q \geq 0$ and $\mathbb{E}[Q] = 2$. Bound $\Pr[Q \geq 8]$ using Markov. Find a distribution attaining the bound.

Full solutions are at the end of the course note.

Practice: Variance and Markov

1. Two independent fair flips: X counts heads. Find $\mathbb{E}[X^2]$, $\mathrm{Var}(X)$, and $\mathrm{Var}(3X - 2)$.
2. Independent X, Y have means 3 and 5, variances 4 and 9. Find the mean and variance of $2X - Y + 1$. Which calculation survives without independence?
3. $Q \geq 0$ and $\mathbb{E}[Q] = 2$. Bound $\Pr[Q \geq 8]$ using Markov. Find a distribution attaining the bound.

3. Bound: $1/4$. Attained by $Q = 8$ with probability $1/4$, and $Q = 0$ otherwise.

Full solutions are at the end of the course note.

Practice: Averages

4. A coin has unknown head probability p. How many independent flips suffice for head-fraction error below 0.01 with probability at least 0.95, using Chebyshev?

5. Let I be the head indicator of one fair flip. Set $X_i = I$ for every i. Find $\mathbb{E}[\overline{X}_n]$ and $\mathrm{Var}(\overline{X}_n)$. Does $\Pr[|\overline{X}_n - 1/2| \geq 1/4]$ tend to zero?

Full solutions are at the end of the course note.